DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 5 (AT2)

Visualization Development Exercise

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT2 — Visualization Development Exercise *Topics: Directory of Visualizations · Visualizing Amounts · Visualizing Distributions*

Question 1 — Chart Selection (Topic 1: Directory of Visualizations)

Task: For each scenario below, name the single best chart type from the directory (Amounts / Distribution / Proportion / x–y Relationship / Geospatial / Network / Temporal / Multivariate) and justify your choice in 1–2 sentences. No code required for this question.

1. Comparing the population of 8 countries in the current year.

Answer:

Showing Comparing the population of 8 countries in the current year.

- **Chart Type:** Bar Chart (Amounts)
- **Justification:** A bar chart clearly compares population values across different countries. It makes it easy to identify the largest and smallest populations.

2. how delivery times vary across 5 courier companies.

Answer:

Showing how delivery times vary across 5 courier companies

- **Chart Type:** Violin Plot (Distribution)
- **Justification:** A violin plot shows the spread, density, and variation of delivery times. It also highlights differences in the distribution between courier companies.

3. Showing the proportion of a household budget spent on rent, food, transport, and savings.

Answer:

Showing the proportion of a household budget spent on rent, food, transport, and savings

- **Chart Type:** Pie Chart (Proportion)
- **Justification:** A pie chart shows how each expense contributes to the total household budget. It is suitable for displaying percentage shares.

4. Showing the relationship between hours studied and exam marks for 50 students.

   Answer:

   Showing the relationship between hours studied and exam marks for 50 students

5. Showing average temperature for each month across 3 different cities at once.

Answer:

Showing average temperature for each month across 3 different cities at once

- Chart Type: Multi-Line Chart (Temporal)
- Justification: A multi-line chart compares temperature changes over time for multiple cities. It clearly shows monthly trends**.**

6. Showing which airports are directly connected by flight routes.

Answer:

- Chart Type: Network Graph (Network)
- Justification: A network graph represents airports as nodes and flight routes as connections. It effectively visualizes connectivity between airports.

Question 2 — Bar Plot (Topic 2: Amounts)

Dataset — app_downloads.csv

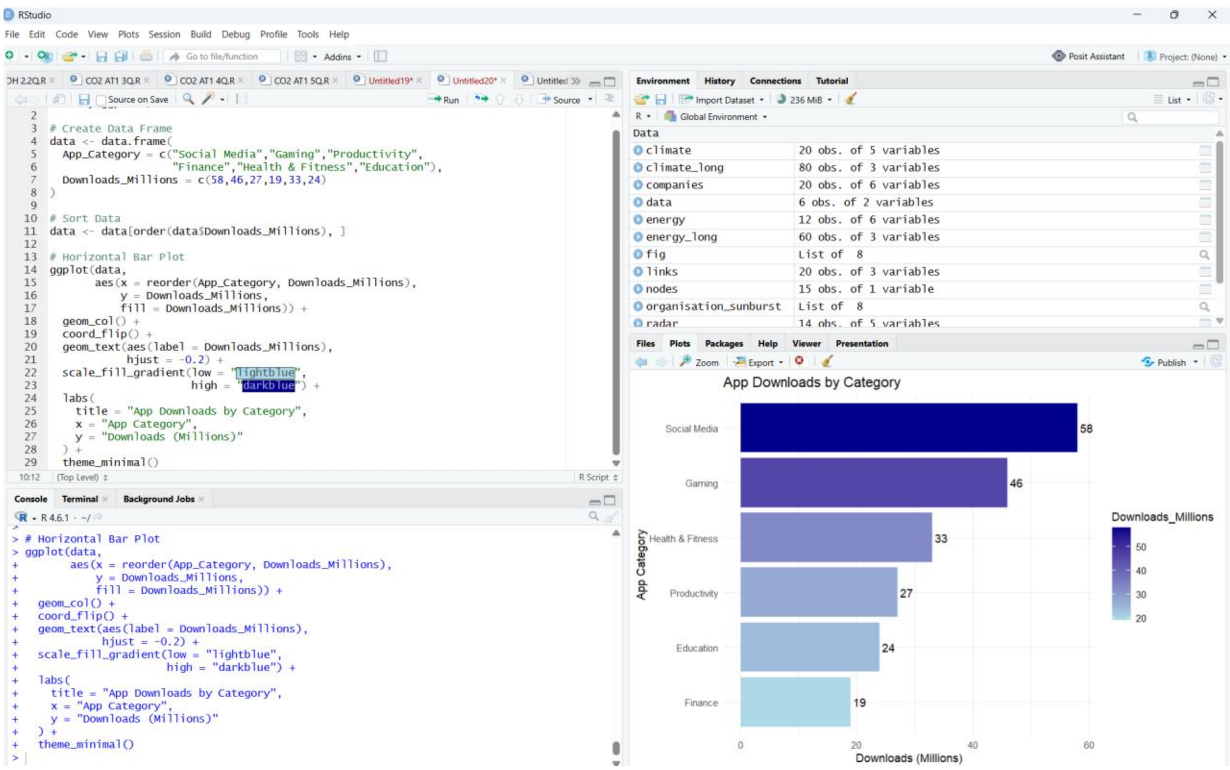
App_Category	**Downloads_Millions**
Social Media	58
Gaming	46
Productivity	27
Finance	19
Health & Fitness	33
Education	24

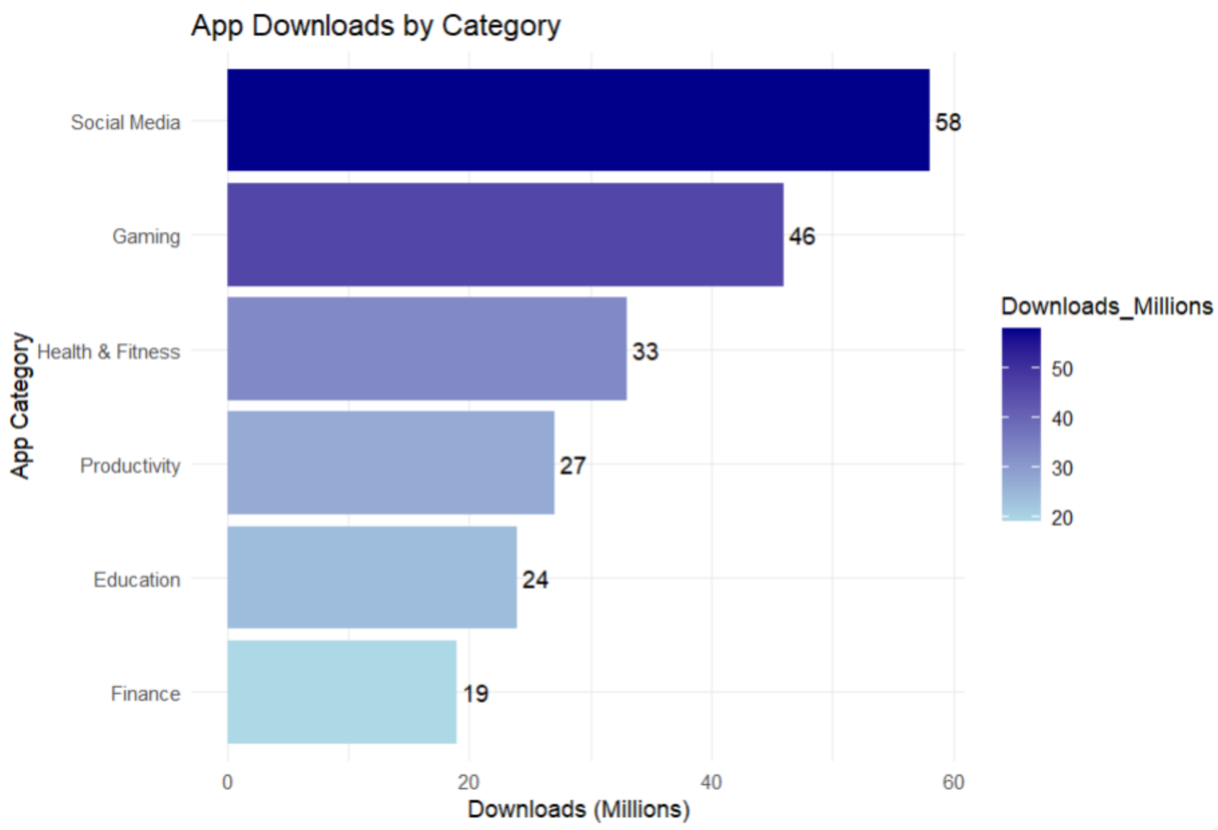
Task:

- Create a horizontal bar plot of Downloads_Millions by App_Category, sorted in ascending order.
- Use a color gradient (single hue, varying shade) so bar color intensity reflects download count.
- Add the exact value as a text label at the end of each bar.

Python: use pandas to build the DataFrame and matplotlib.pyplot.barh (or seaborn.barplot with orient="h").

R: use a data.frame, ggplot2::geom_col() with coord_flip() (or geom_col() plus aes(y = reorder(...))).





Question 3 — Grouped/Stacked Bars, Dot Plot & Heatmap (Topic 2: Amounts)

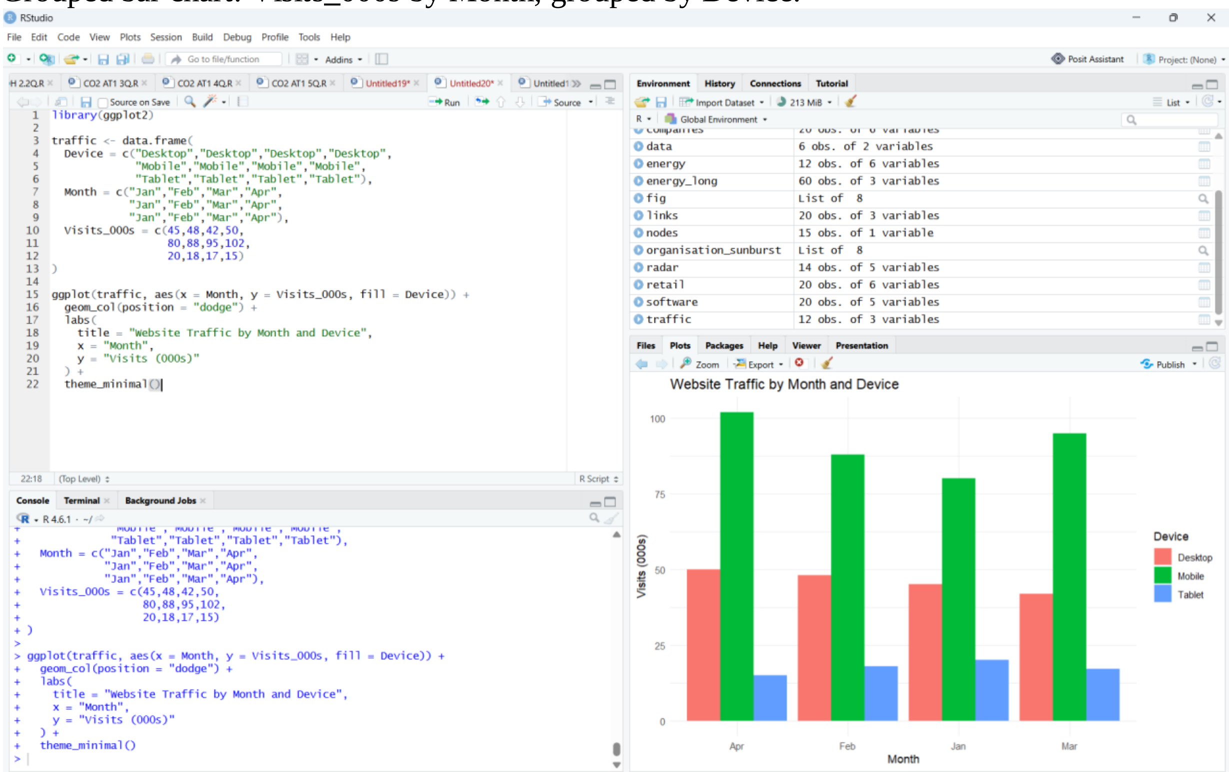
Dataset — website_traffic.csv

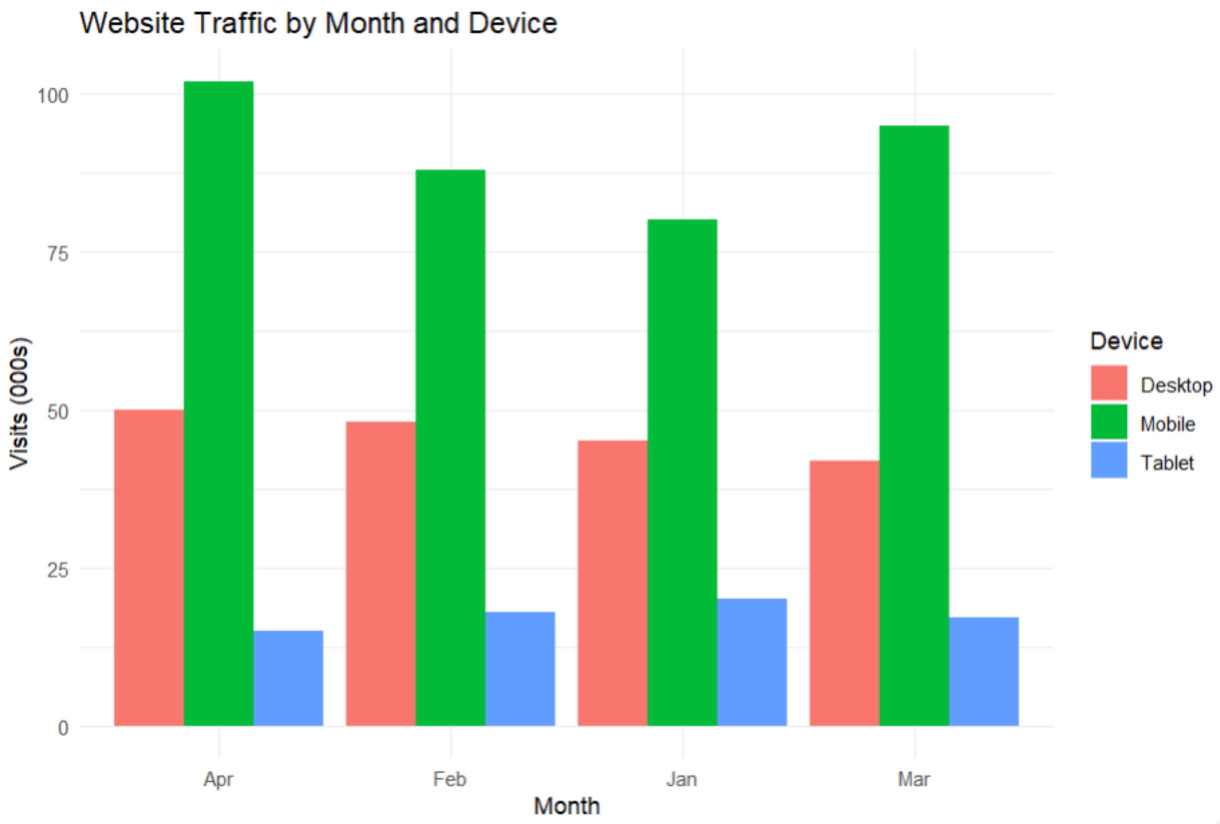
Device	Month	Visits_000s
Desktop	Jan	45
Desktop	Feb	48
Desktop	Mar	42
Desktop	Apr	50
Mobile	Jan	80
Mobile	Feb	88
Mobile	Mar	95
Mobile	Apr	102

Tablet	Jan	20
Tablet	Feb	18
Tablet	Mar	17
Tablet	Apr	15

Task (produce all four views from the same dataset):

7. Grouped bar chart: Visits_000s by Month, grouped by Device.

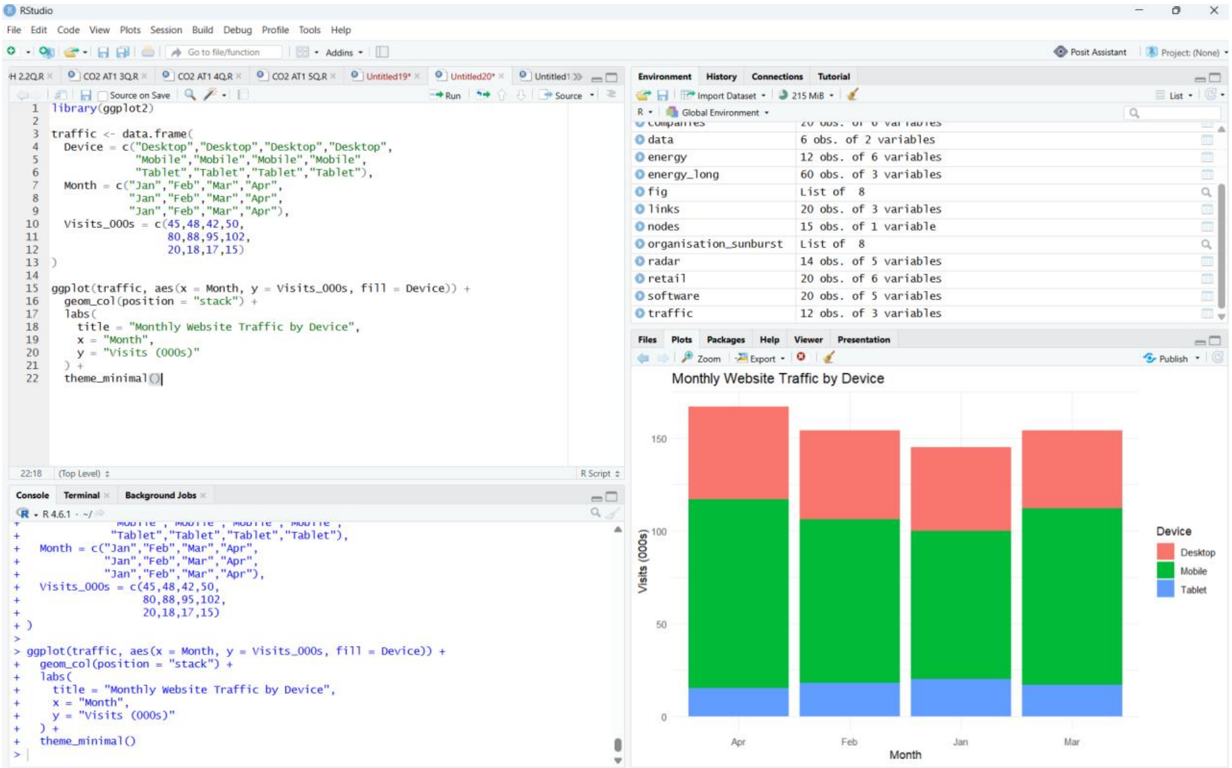


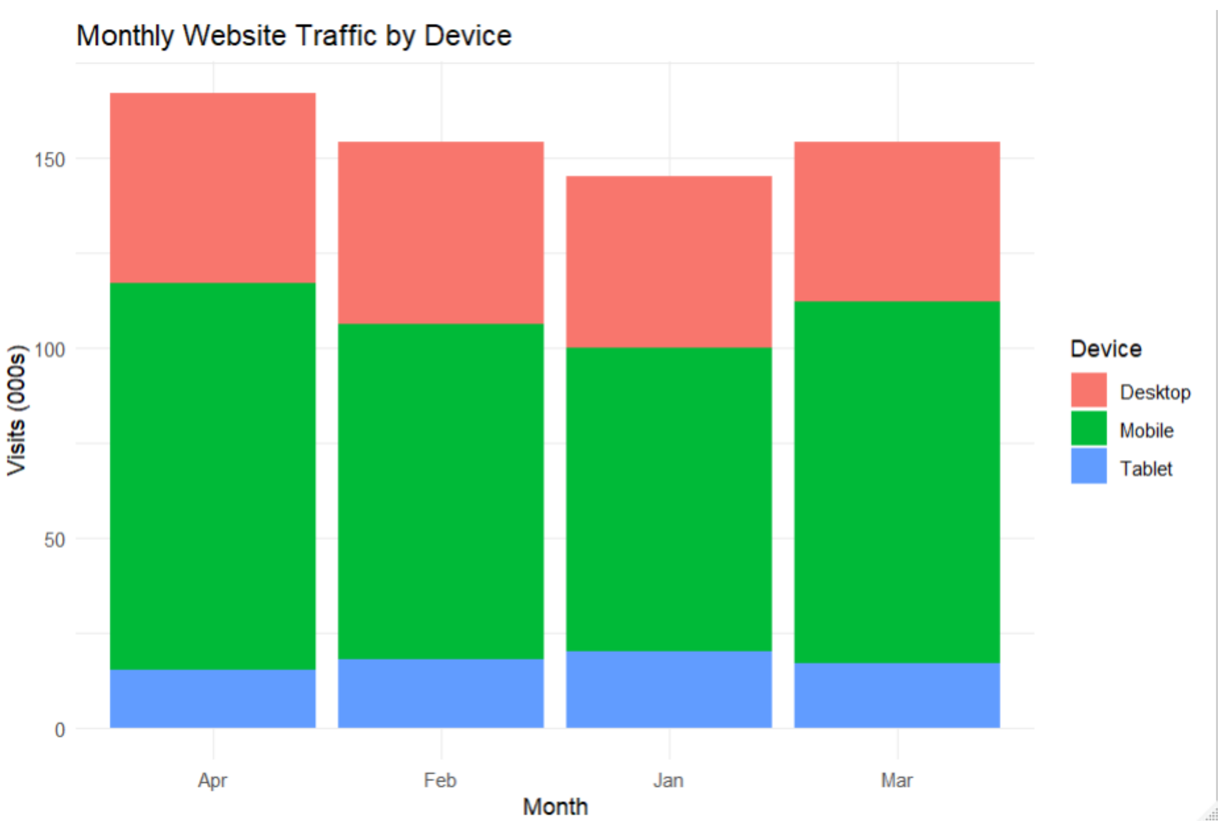


Interpretation

- Mobile devices recorded the highest website traffic in every month.
- Desktop traffic remained moderate, while Tablet traffic was the lowest.
- The grouped bar chart makes it easy to compare traffic across devices for each month.

8. Stacked bar chart: same data, stacked by Device within each Month, so each bar shows total monthly traffic.





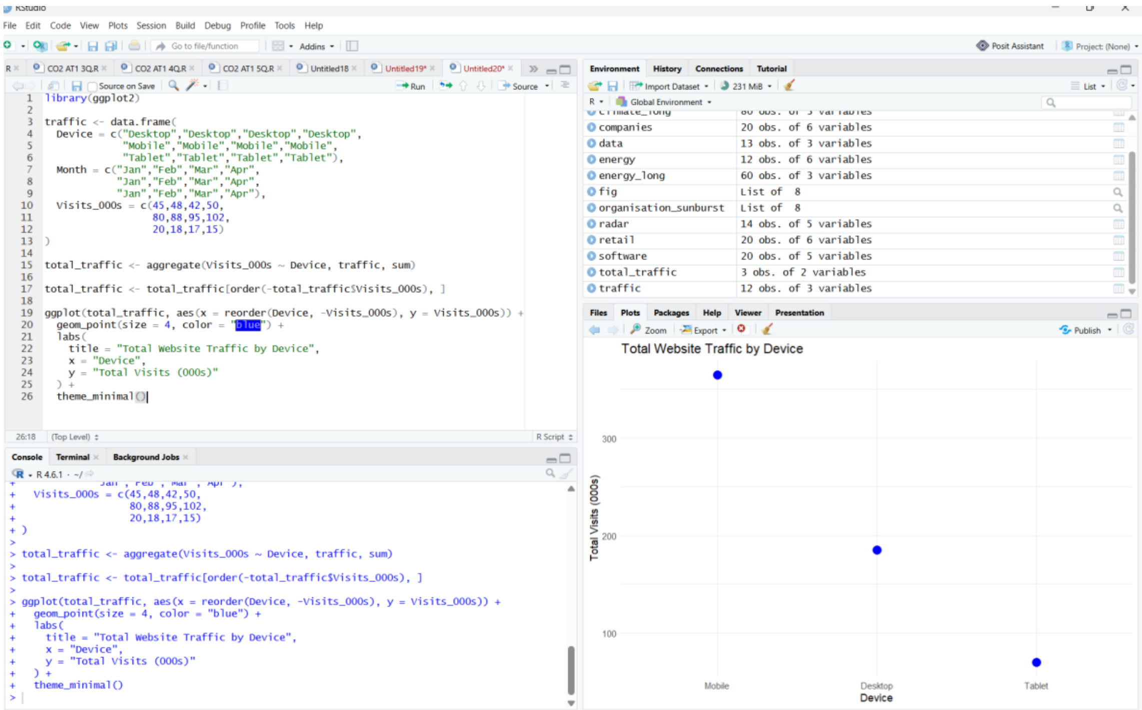
Interpretation:

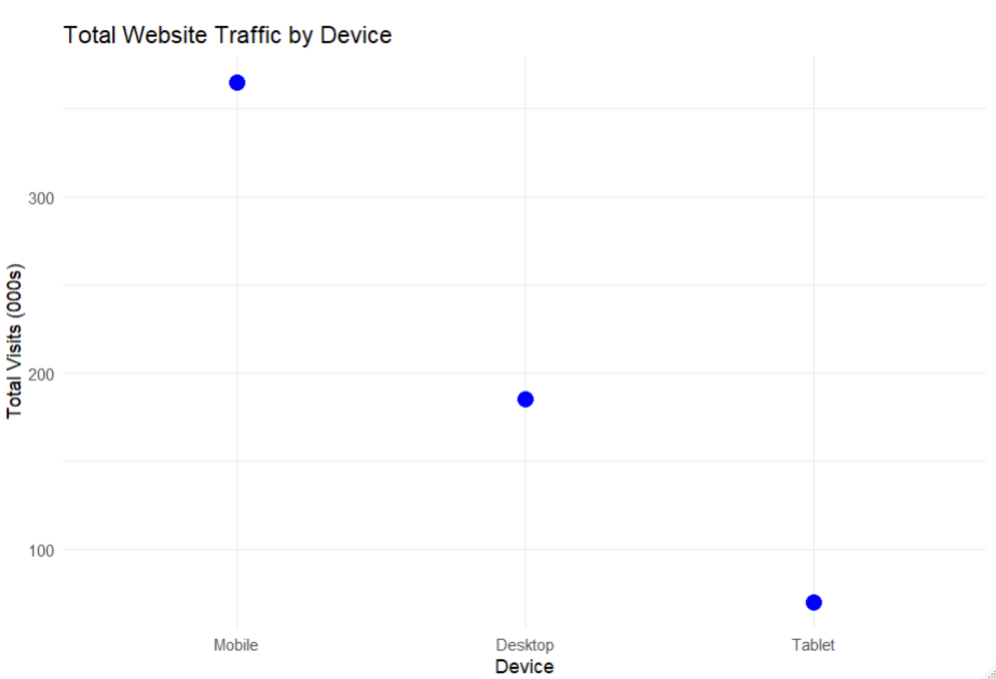
- The total website traffic increases from January to April.

- Mobile contributes the largest share of traffic each month.
- Tablet contributes the smallest share in every month.

9. Dot plot: total 4-month traffic per device (sum across months), ranked from highest to lowest.

Answer:





Interpretation:

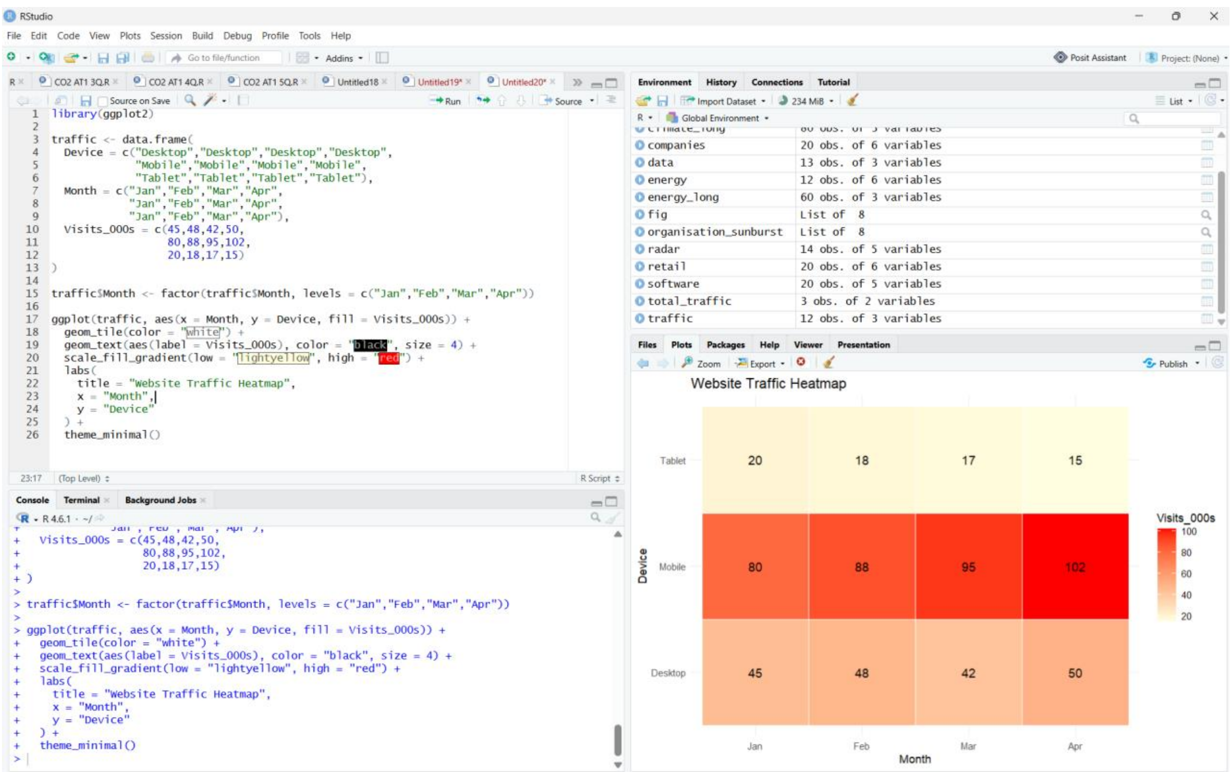
- Mobile has the highest total website traffic over the four months.
- Desktop has moderate traffic, while Tablet has the lowest.
- The dot plot clearly ranks devices based on total visits.

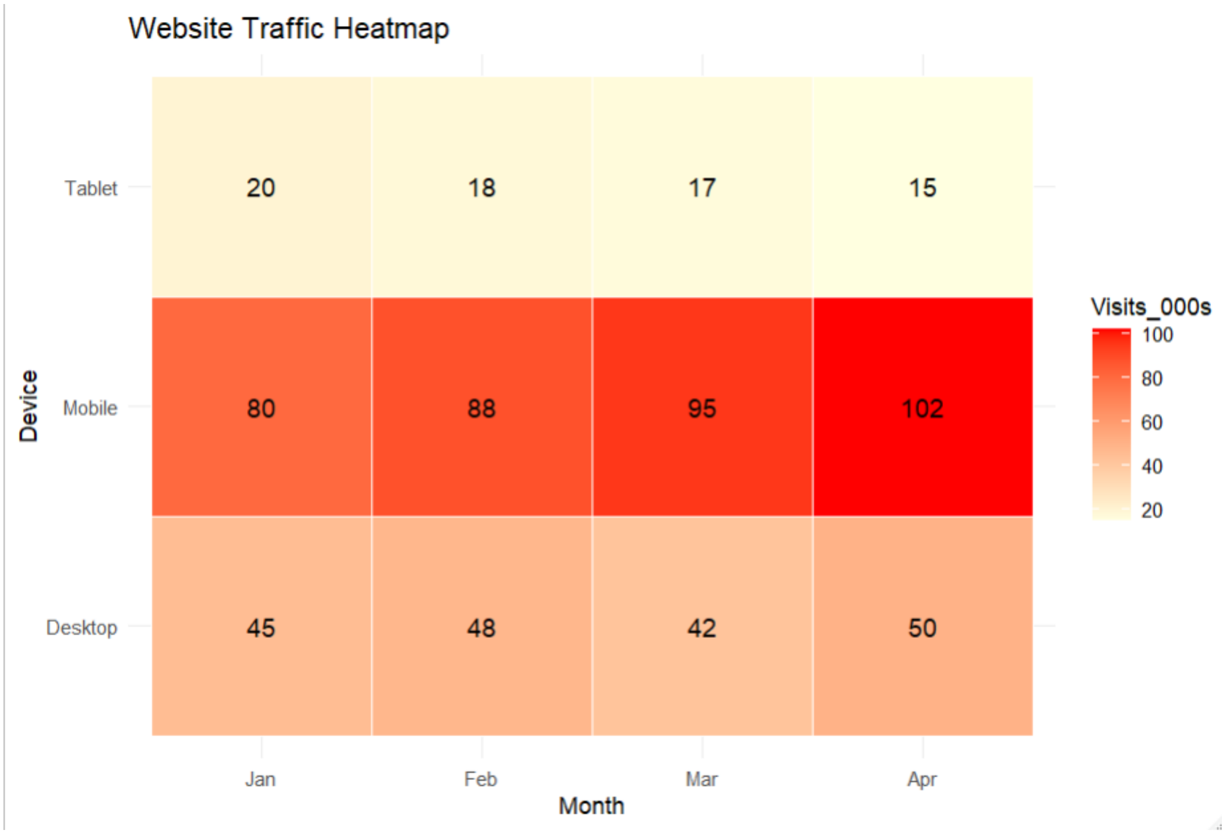
10. Heatmap: Device (rows) × Month (columns), color-encoded by Visits_000s.

Python: pandas.pivot_table to reshape; matplotlib/seaborn for grouped/stacked bars, seaborn.heatmap for the heatmap, matplotlib.pyplot.scatter (or plt.plot with markers) for the dot plot.

R: tidyr::pivot_wider where needed; ggplot2::geom_col(position = "dodge") for grouped, position = "stack" for stacked, geom_point() for the dot plot, geom_tile() for the heatmap.

Answer:





Interpretation:

- Darker colors represent higher website traffic values.
- Mobile recorded the highest traffic, especially in April.
- Tablet recorded the lowest traffic across all months.

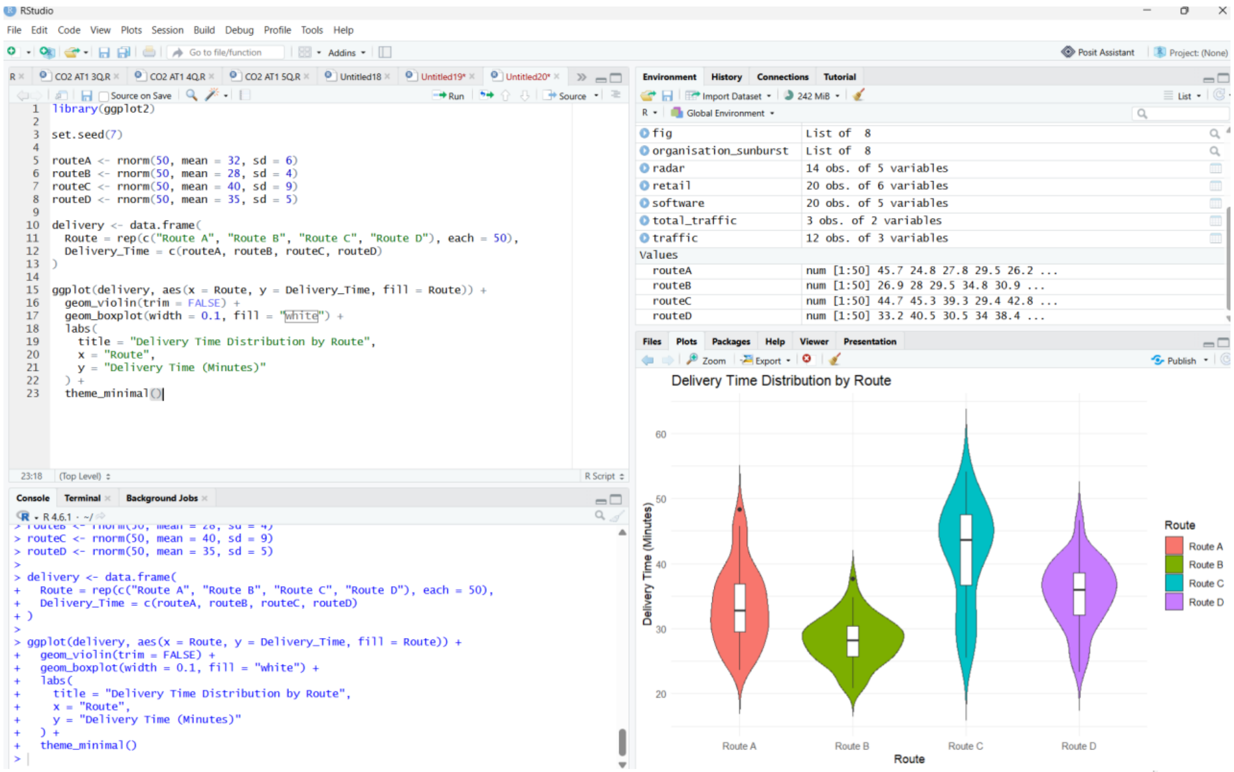
Question 4 — Violin & Ridgeline Plots (Topic 3: Distributions) **Dataset** — generate synthetically for reproducibility:

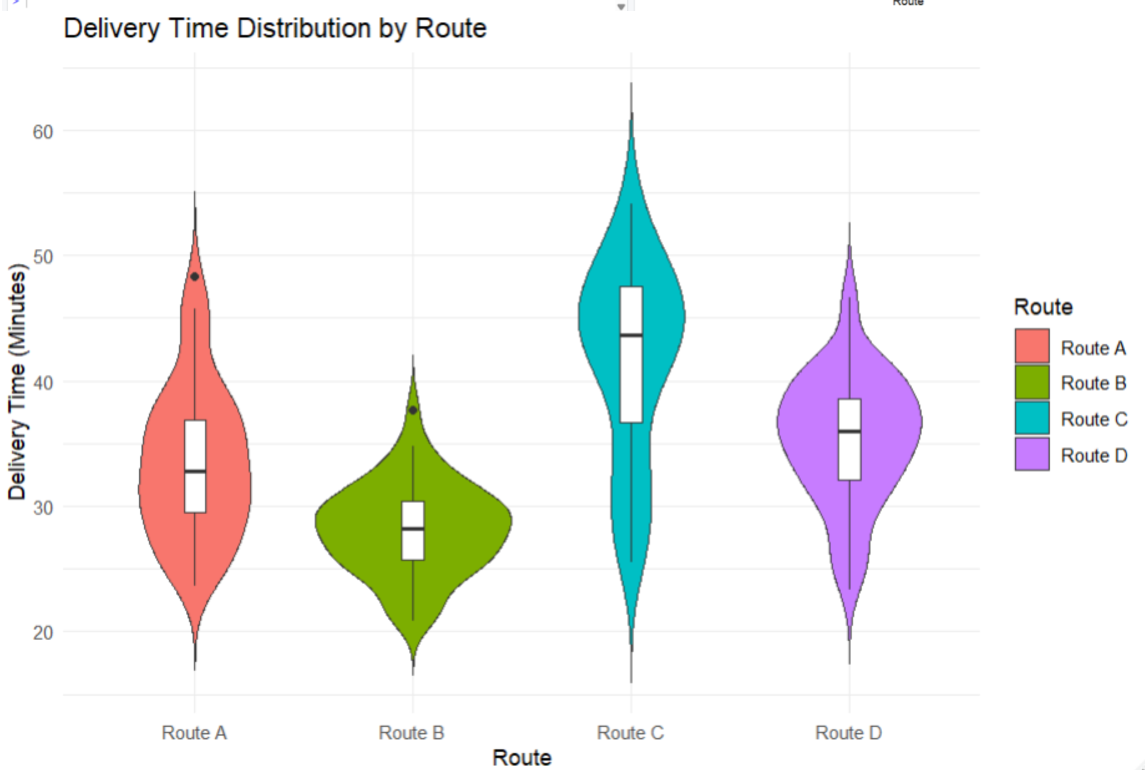
```
Route A: 50 delivery times (minutes) ~ Normal(mean=32, sd=6)
Route B: 50 delivery times (minutes) ~ Normal(mean=28, sd=4)
Route C: 50 delivery times (minutes) ~ Normal(mean=40, sd=9)
Route D: 50 delivery times (minutes) ~ Normal(mean=35, sd=5)
```

(Set a random seed of 7 in both Python and R so results are consistent across languages.) **Task:**

11. Violin plot: delivery-time distribution by route (all 4 routes on one plot), with an embedded boxplot or median marker.

Answer:





Interpretation:

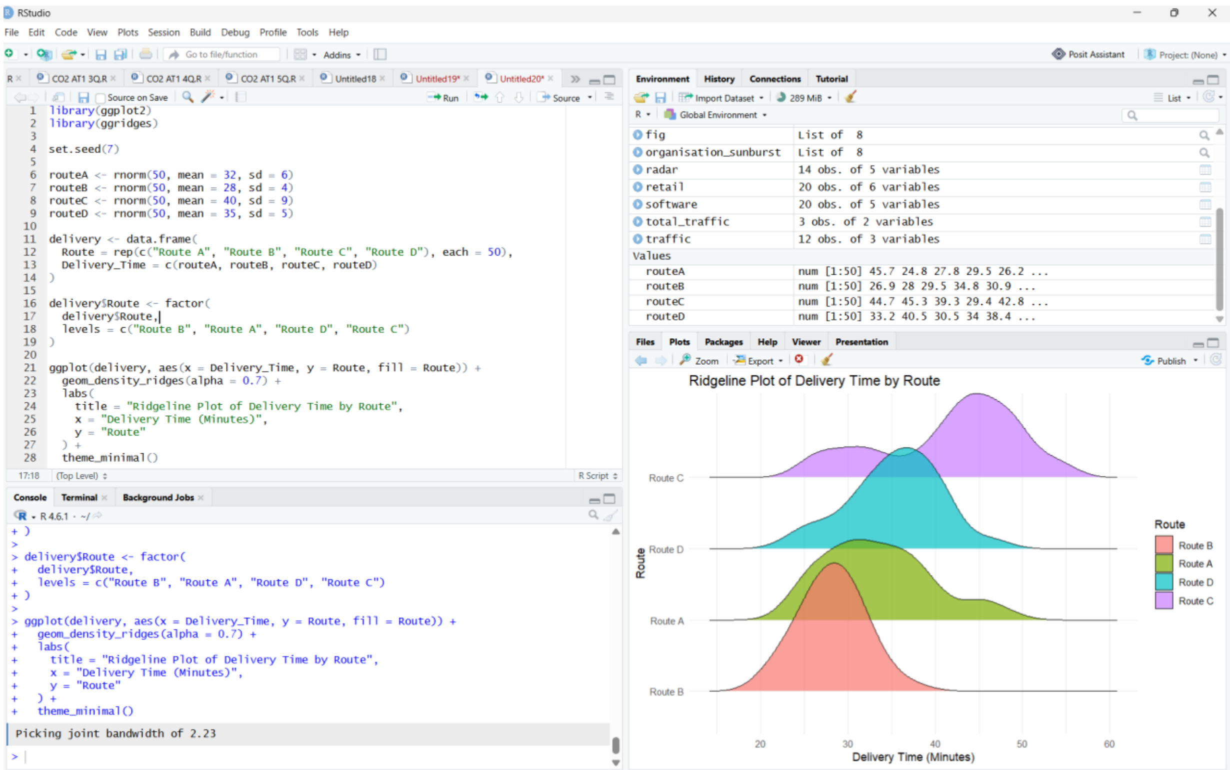
- Route B has the shortest average delivery time.
- Route C shows the greatest variation in delivery times.
- The violin plot with an embedded boxplot clearly compares the distribution and median of all four routes.

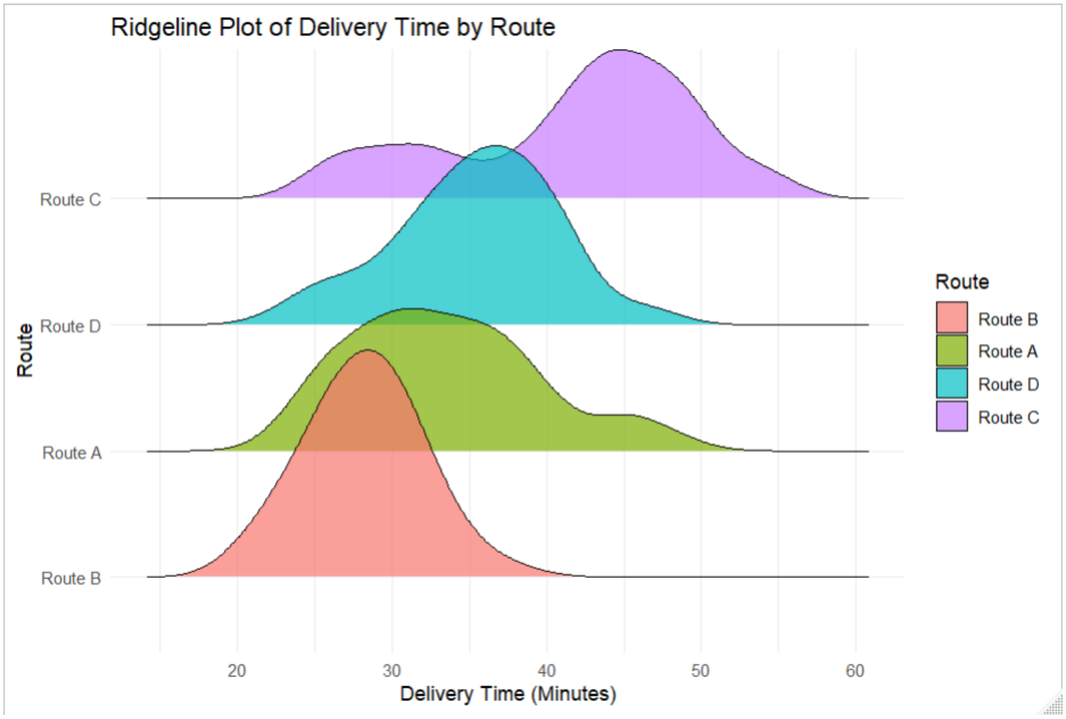
12. Ridgeline plot: same 4 routes, overlapping density curves, ordered by mean delivery time (fastest at the bottom).

Python: numpy.random.normal to generate data, seaborn.violinplot for the violin plot, joypy.joyplot (or a stacked seaborn/matplotlib kdeplot FacetGrid) for the ridgeline.

R: rnorm() to generate data, ggplot2::geom_violin() for the violin plot, ggridges::geom_density_ridges() for the ridgeline.

Answer:





Interpretation:

- Route B appears at the bottom because it has the lowest average delivery time.
- Route C appears at the top because it has the highest average delivery time.
- The ridgeline plot compares the distribution of delivery times across all four routes, making differences in spread and density easy to observe.

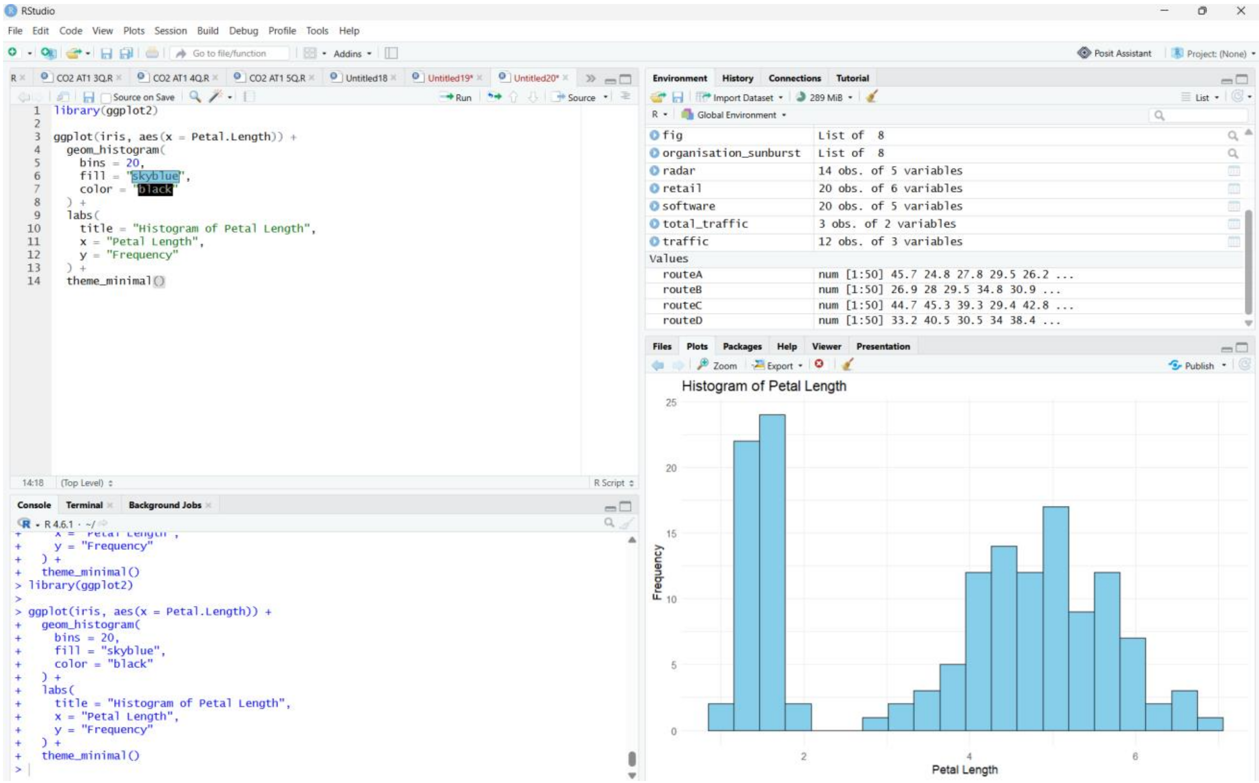
Question 5 — Histograms & Density Plots (Topic 3: Distributions) **Dataset:** Use a built-in dataset available in both languages.

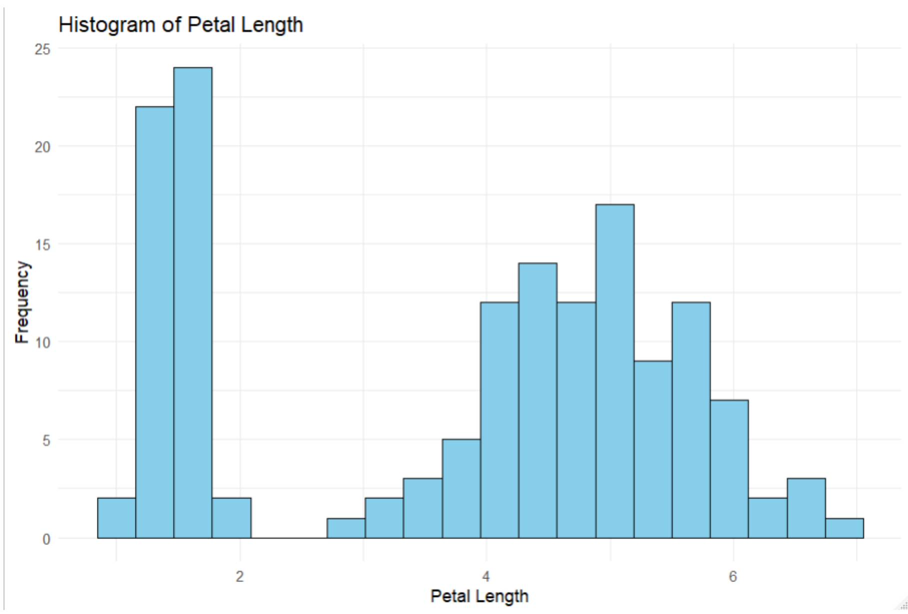
- Python: seaborn.load_dataset("iris") → use the petal_length column.
- R: the built-in iris dataset → use the Petal.Length column.

Task:

13. Plot a histogram of the chosen continuous variable with 20 bins.

Answer:



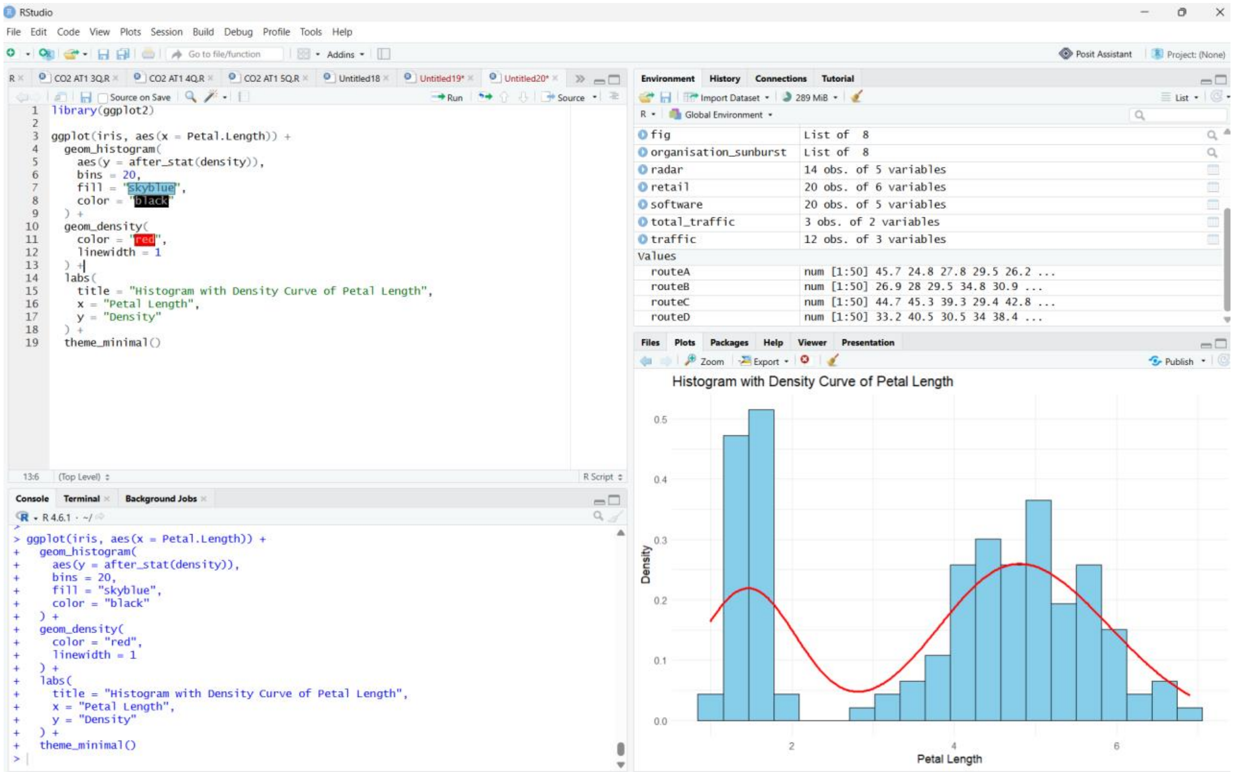


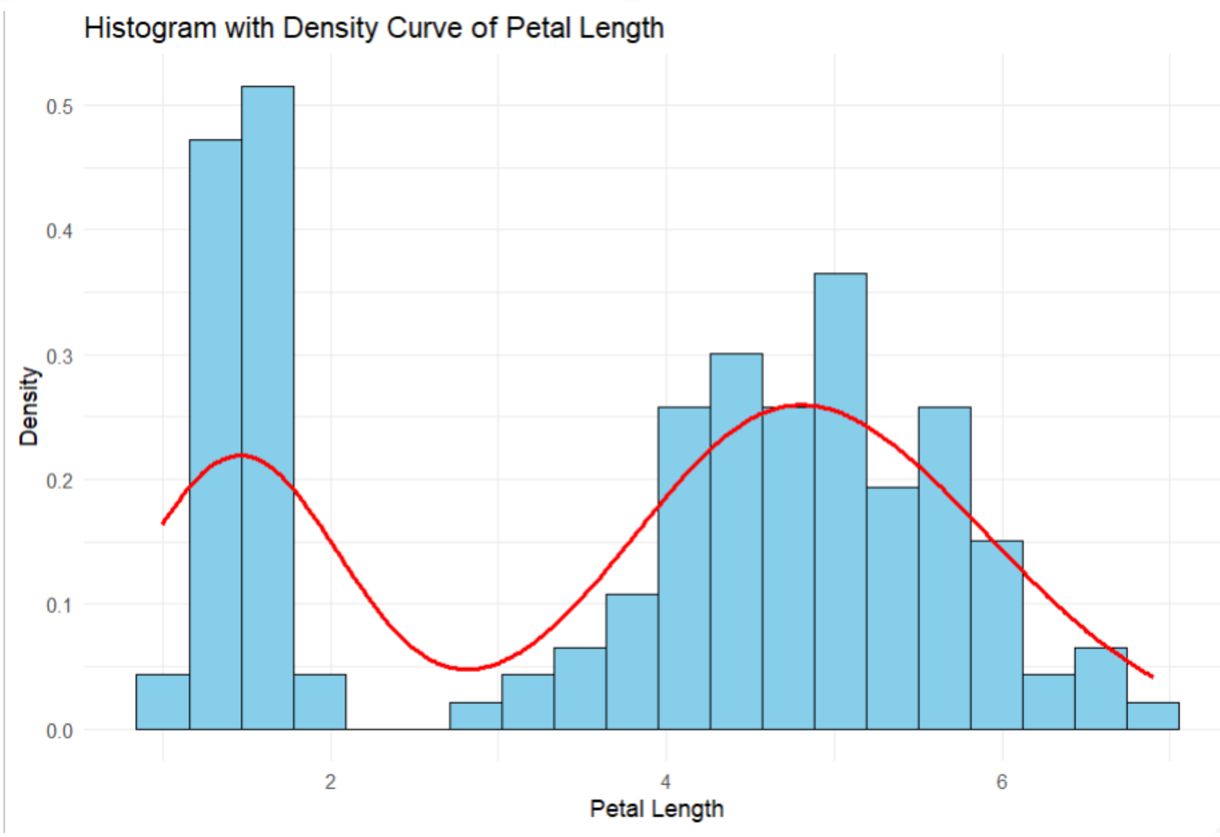
Interpretation:

- The histogram shows the frequency distribution of petal lengths in the iris dataset.
- Most observations are concentrated in the middle range of petal lengths.
- The chart helps identify the overall distribution of the data.

14. Overlay a kernel density estimate (KDE) curve on the same plot.

Answer:





Interpretation:

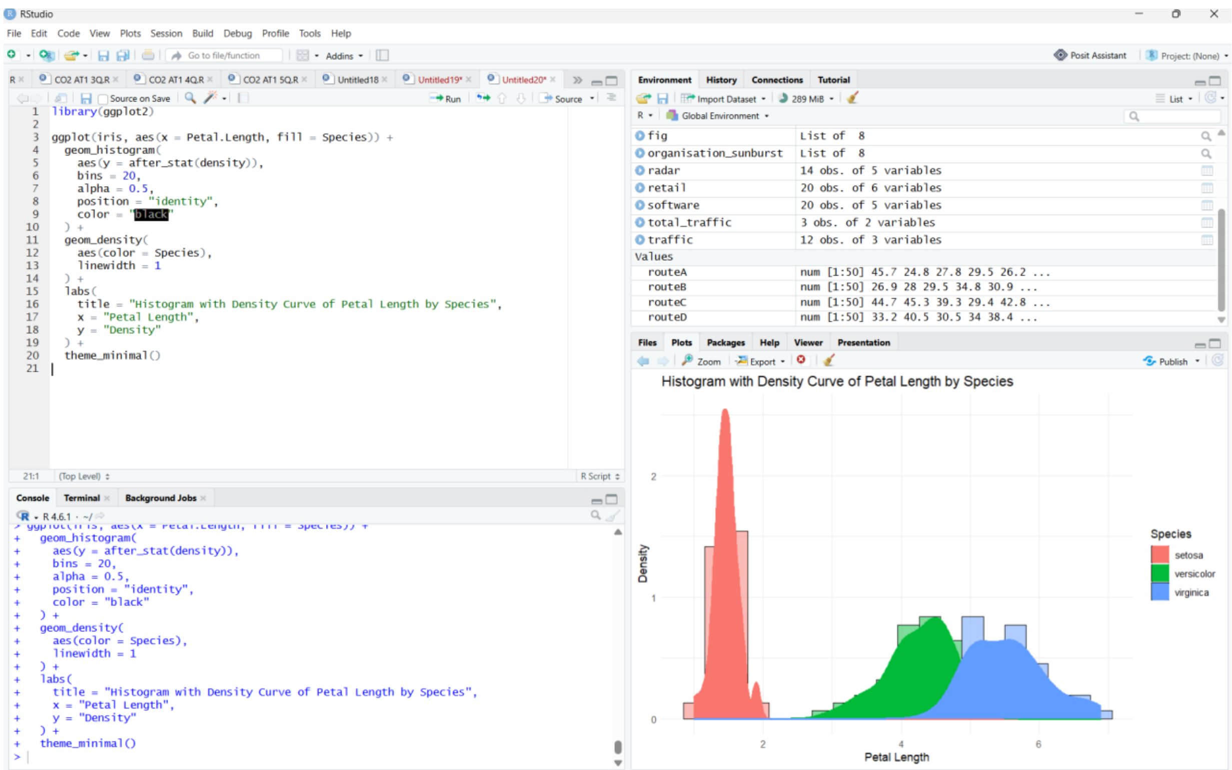
- The histogram displays the distribution of petal lengths in the iris dataset.
- The density (KDE) curve provides a smooth representation of the data distribution.
- Most petal length values are concentrated around the central region.

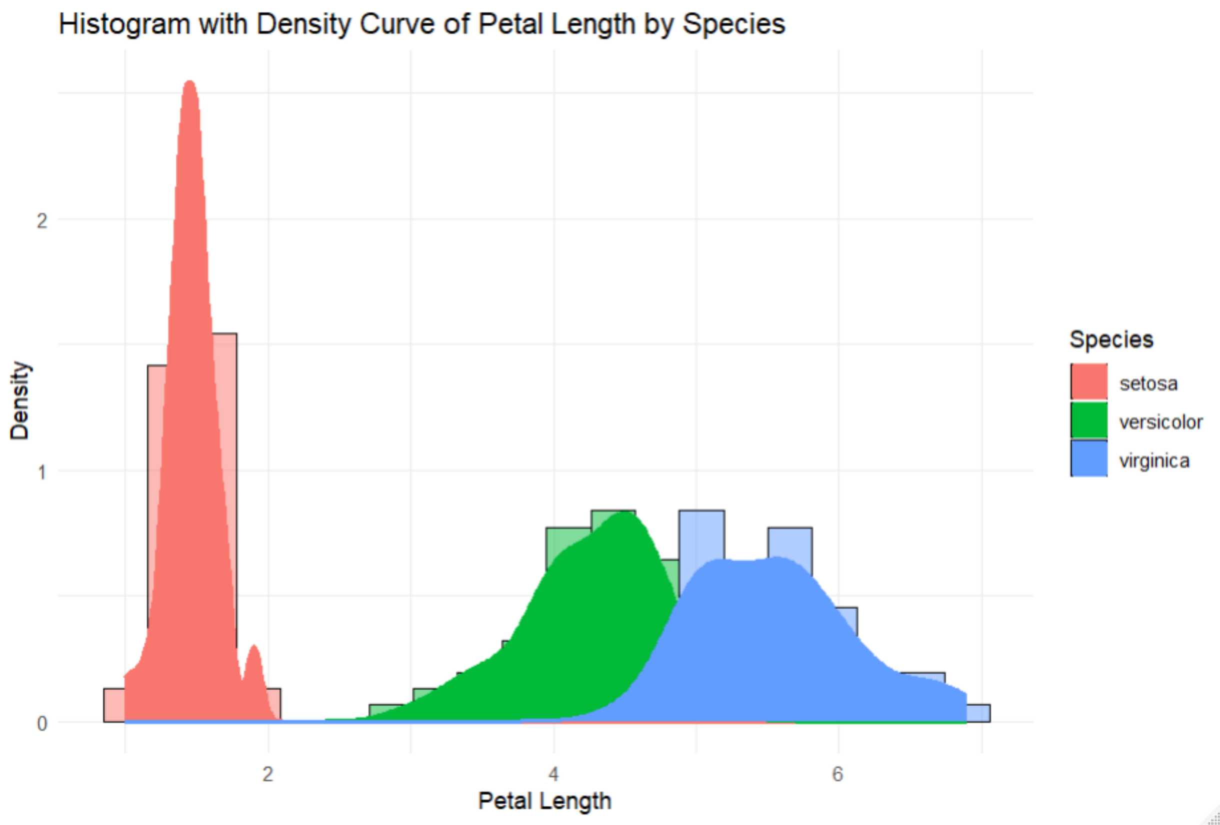
15. Repeat both, split/colored by species (setosa, versicolor, virginica), to compare distribution shape across groups.

Python: seaborn.histplot(..., kde=True) or matplotlib.pyplot.hist + scipy.stats.gaussian_kde.

R: ggplot2::geom_histogram() combined with geom_density() (use after_stat(density) so scales match), with fill = mapped to Species for the grouped version.

Answer:





Interpretation:

- **Setosa** has the smallest petal lengths among the three species.
- **Versicolor** has medium petal lengths, while **Virginica** has the largest petal lengths.
- The grouped histogram and density curves clearly compare the distribution of petal lengths for each species.

Submission Checklist

- Q1: written chart-type justifications (no code)
- Q2: bar plot — Python + R
- Q3: grouped bar, stacked bar, dot plot, heatmap — Python + R
- Q4: violin plot, ridgeline plot — Python + R
- Q5: histogram + density (overall and grouped) — Python + R
- Each chart has a title, axis labels, and a short written interpretation